Internet Crimes and Cyber Security

Week: 05 **Session: 21**

Topic :Introduction to tools and techniques to Counter Cyber Attacks using VAPT Services

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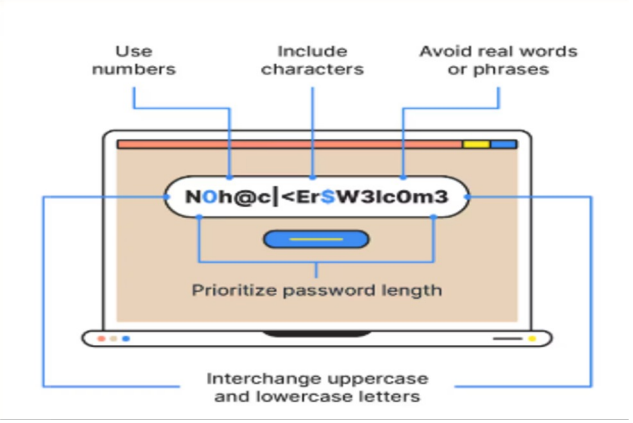
Tools and techniques to Counter Cyber Attacks

Concepts to be covered

1. Basic measures to Counter Cyber Threats and Attacks
2. What are VAPT services
3. Countering Cyber Threats technically, using Vulnerability Assessment Tools
4. Countering Cyber Attacks technically, using Penetration Testing Tools
5. Understanding Incidence Response and Recovery Platforms

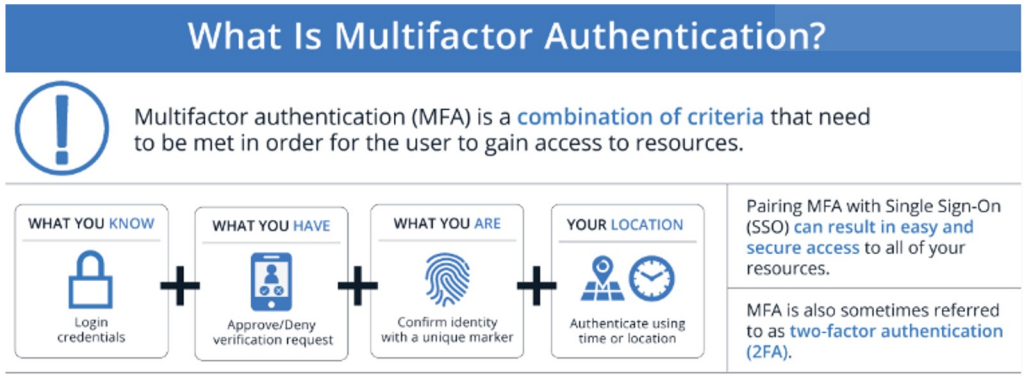
Tools and techniques to Counter Cyber Attacks

1. Basic measures to Counter Cyber Threats and Attacks



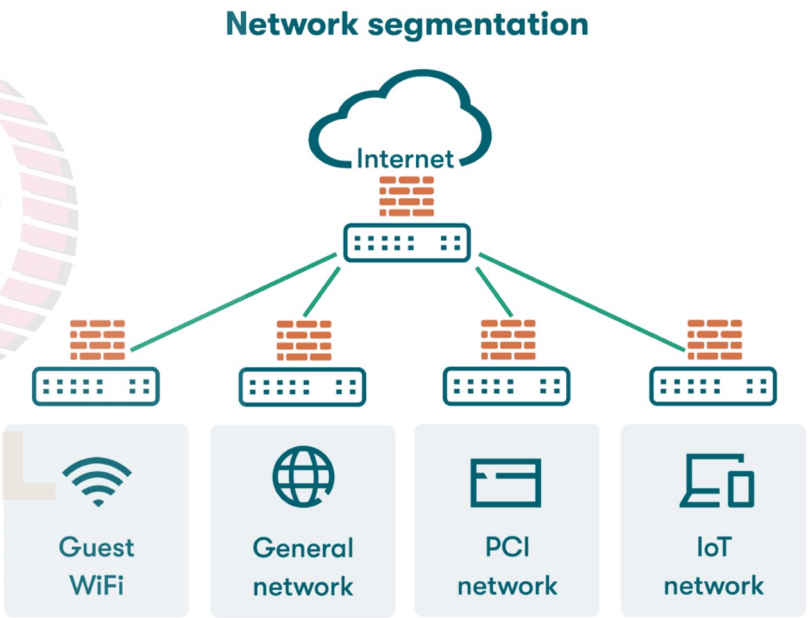
- **Strong Password Policies**
  Use long, complex, and unique passwords

- **Multi-Factor Authentication (MFA)**
  Adds extra verification (OTP, biometrics)
  Fixes known security vulnerabilities



Tools and techniques to Counter Cyber Attacks

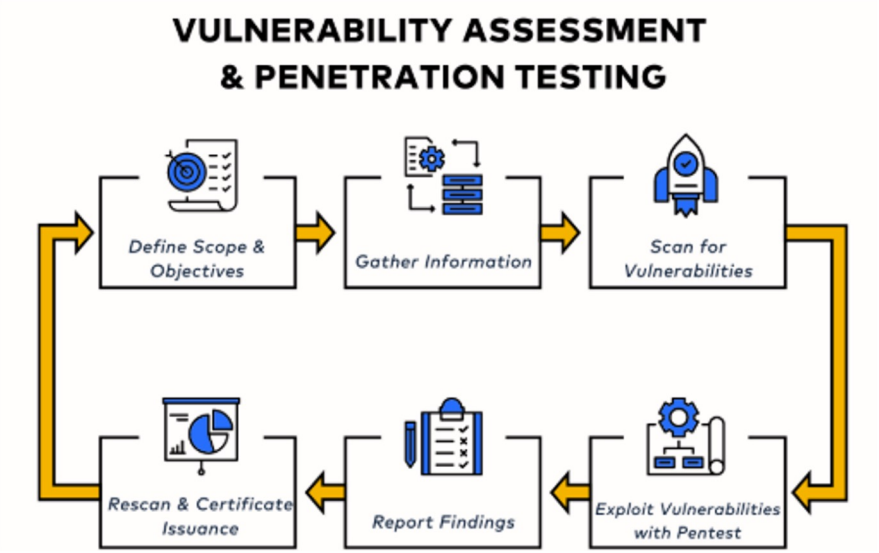
- **Network Segmentation**
  Limits attack spread within a network
- **User Awareness & Training**
  Prevents phishing and social engineering atta
- **Access Control & Least Privilege**
  Users get only necessary permissions
- **Regular Security Audits & Penetration Testing**
  Identifies weaknesses before attackers do



Tools and techniques to Counter Cyber Attacks

2. What are VAPT services

VAPT stands for Vulnerability Assessment and Penetration Testing, a crucial cybersecurity practice **where organizations find and fix security weaknesses in their systems (networks, apps, cloud) by combining automated vulnerability scanning (VA) with simulated real-world attacks i.e Penetration Testing (PT)** by Cyber Security Professionals to see how bad Threat actors could exploit them, thus strengthening overall digital defenses



- Vulnerability Assessment assists in locating known system, network, or application weaknesses or misconfigurations.
- Penetration Testing makes use of those weaknesses safely to measure the actual consequences of the same in the real world, and how far an attacker can go.

Tools and techniques to Counter Cyber Attacks

3. Countering Cyber Threats technically using Vulnerability Assessment Tools

There are several types of vulnerability assessments. These include:

1.Host assessment – The assessment of critical servers, which may be vulnerable to attacks if not adequately tested or not generated from a tested machine image.
2.Network and wireless assessment – The assessment of policies and practices to prevent unauthorized access to private or public networks and network-accessible resources.
3.Database assessment – The assessment of databases or big data systems for vulnerabilities and misconfigurations, identifying rogue databases or insecure dev/test environments, and classifying sensitive data across an organization's infrastructure.
4.Application scans – The identifying of security vulnerabilities in web applications and their source code by automated scans on the front-end or static/dynamic analysis of source code.

Tools and techniques to Counter Cyber Attacks

1.Web application scanners that test for and simulate known Threat

Tools are :

1. https://urlscan.io/
2. https://www.sitelock.com/free-website-scan/
3. https://www.ssllabs.com/ssltest/
4. https://web-check.xyz/

2. **Protocol scanners** that search for vulnerable protocols, ports and network services.

Tools are :

1. https://dnschecker.org/all-tools.php
2. https://scan.cyberchief.ai/

3. **Network scanners** that help visualize networks and discover warning signals like stray IP addresses, spoofed packets and suspicious packet generation from a single IP address.

Thank You